converted into a usable output is called secondary transducers.

iii. Passive and active transducers

The transducers which are externally powered are called passive transducers and self-generating types are called as active transducers.

iv. Analog and digital transducers

The transducer which gives output in analog form is called analog transducers and transducer which gives output in digital form is called digital transducers.

v. Transducers and inverse transducers

The transducers which converts non-electrical quantity into electrical quantity is called transducers and the transducers which converts electrical quantity into non-electrical quantity is called inverse transducers.

7.4 Factors to be considered for the selection of transducers

The selection of transducers is based on the applications where it is used. The following important factors are to be considered for the selection of transducers.

i. It should have high input impedance and low output impedance, to avoid loading effect.
ii. It should have good resolution over is entire selected range.
iii. It must be highly sensitive to desired signal and insensitive to unwanted signal.
iv. Preferably small in size.
v. It should be able to work in corrosive environment.
vi. It should be able to withstand pressure, shocks, vibrations etc...
vii. It must have high degree of accuracy and repeatability.
viii. Selected transducer must be free from errors.

Do you Know?

A **biometric device** is a security identification and authentication device. Such devices use automated methods of verifying or recognizing the identity of a living person based on a physiological or behavioral characteristic. These characteristics include finger prints, facial images, iris and voice recognition. Biometrics are being used to establish better and accessible records of the hours employee's work



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